

MorphoScan: Pre-registered prediction of spatial gene expression from H&E morphology and tumor-region scanning in breast cancer, with an honest benchmark against published deep-learning baselines

MorphoScan pipeline (MEGA27 item 17-spatial)

September 25, 2026

Abstract

Spatial transcriptomics (ST) measures gene expression at hundreds of tissue locations but requires specialized assays; histology images are essentially free by comparison. MorphoScan asks a narrow, falsifiable question: how much spatial expression signal is recoverable from hematoxylin-and-eosin (H&E) morphology alone with a fully interpretable feature set, and where does that honest number sit relative to published deep-learning systems? We pre-registered every gate before seeing results (PREREG.md, committed first). On the public 68-section, 23-patient breast-cancer ST cohort (Mendeley 29ntw7sh4r v5), using leave-one-patient-out evaluation on the ten HER2-positive patients and a frozen cross-subtype transport cohort, a ridge probe over 41 handcrafted morphology features reaches a median per-gene held-out Pearson correlation of TBD across 250 genes (permutation p TBD). A Gaussian-kernel spatial smoothing stage (the MorphoScan method advance) shifts this to TBD (paired Wilcoxon p TBD). We report the comparison to the published ST-Net baseline (median $r \approx 0.19$) and HisToGene-class systems exactly as measured, including where we do not beat them. A companion tumor-region scanner reaches patient-held-out AUC TBD (logistic) / TBD (gradient boosting). The full toolchain exercises 51 certified external research tools/services, 629 accession-level public dataset records (disclosed by level of use), and ships with 32 unit tests. All negatives are preserved.

1 Introduction

1.1 Why morphology-to-expression prediction matters

Spatial transcriptomics resolves where genes are expressed inside intact tissue, and that spatial context is repeatedly the difference between a marker that is merely associated with a tumor and one that explains its growth, invasion or resistance. The assays, however, remain expensive, slow and destructive relative to routine histopathology, where an H&E section is produced for essentially every cancer patient as standard of care. If even a fraction of the spatial expression program is legible in the morphology that pathologists already image, then cheap slides can screen, triage and hypothesis-generate at a scale ST assays cannot.

This is not a new idea. ST-Net (He et al., Nature Biomedical Engineering, 2020) trained a DenseNet-121 to predict the expression of 250 genes per spot from image patches on the HER2-positive subset of the same cohort we use, reporting a median per-gene Pearson correlation near 0.19 on held-out patients. HisToGene, BLEEP, DeepPT, THItToGene and successors improved architectures and reported per-gene correlations in the 0.2–0.4 range on related benchmarks. What

the literature is quieter about is three things we care about here: (i) how much of that signal is already carried by simple, inspectable morphology statistics; (ii) how fragile the signal is when the evaluation is honestly patient-held-out rather than spot-held-out; and (iii) how far predictions transport across tumor subtypes and platforms.

1.2 What this project claims, and what it does not

MorphoScan is deliberately built on interpretable features — color-deconvolution stain concentrations, gray-level co-occurrence textures, entropy and color moments — so that every predictor weight maps to a morphology a pathologist can name. We do not claim to beat deep networks at their own game; where published systems outperform our probe we say so, in numbers. We do claim: (a) a pre-registered, leakage-free estimate of morphology-expression coupling on a standard cohort; (b) a simple, principled method advance — spatial graph smoothing of predictions — that exploits spatial autocorrelation the per-spot models ignore, evaluated against its own locked gate; (c) an honest transport experiment to non-HER2 subtypes; and (d) a working, tested, documented tool that performs both expression prediction and tumor-region scanning end to end.

1.3 The scanning problem

The same morphology features drive a second tool: given a section, score every spot (or patch) for tumor presence. Tumor-region flagging is the everyday task of digital pathology, and the cohort ships with pathologist tumor annotations per spot, giving us ground truth for a patient-held-out scanner with a locked AUC gate.

2 Data

2.1 Primary cohort

All primary analysis uses Human breast cancer in situ capturing transcriptomics (Mendeley Data 29ntw7sh4r, version 5; Stenbeck, Bergenstråhle, Lundeberg, Borg and colleagues). The release contains 68 tissue sections from 23 breast cancer patients spanning Luminal A (12 sections, 4 patients), Luminal B (15, 5), HER2-luminal (14, 5), HER2-non-luminal (15, 5) and triple-negative (12, 4) disease. Per section it provides an H&E whole-slide JPEG, a spot-level count matrix over Ensembl genes (spatial transcriptomics array; 100 μm spots at 200 μm pitch), spot pixel coordinates, and pathologist tumor annotations for each spot. Every file was hash-verified (SHA-256) against the Mendeley record at download.

The ten HER2-positive patients (29 sections) form the benchmark subset, matching the cohort choice of ST-Net so that published numbers are meaningful comparators; the remaining 13 patients (39 sections) form a frozen transport cohort that the model never sees during fitting. Genes are aligned across sections by Ensembl identifier; the cross-section intersection spans TBD genes, and per-section library sizes are normalized to 10^4 counts followed by $\log(1+x)$.

2.2 Registry harvest (dataset ledger)

Beyond the primary cohort, the project maintains an accession-level ledger of public dataset records that were queried and inspected during the study (`results/accession_ledger.csv`). The ledger currently holds 629 records: 69 analyzed-primary records (the 68 sections plus the cohort record), and metadata-mined records from GEO DataSets (400 series matching spatial

transcriptomics), the NCI Genomic Data Commons (80 projects), CELLxGENE (40 collections) and The Cancer Imaging Archive (40 collections). Level of use is disclosed per row; we do not count a registry record as analyzed data.

3 Methods

3.1 Pre-registration and evaluation discipline

Every gate in this study was locked in `PREREG.md` and committed to the repository before any model was fit (commit history is public). Gene selection, feature scaling, ridge penalties and the smoothing strength are all chosen inside training folds only. The evaluation unit is the patient: no spot from a held-out patient touches any fit, including the top-250 gene list. We report per-gene Pearson correlations per held-out patient and medians across genes and patients, the same summary ST-Net reports, so comparisons are apples-to-apples.

3.2 Morphology features

Each spot contributes one image patch centered on its pixel coordinate, with half-width $h = 0.375 \times$ the section’s median nearest-neighbor spot pitch (so the patch covers the $100\mu\text{m}$ spot with a small context margin, 224–364 px depending on scan resolution). From each patch we compute 41 features:

Stain concentrations (Beer–Lambert deconvolution). Optical density is $\text{OD}_c = -\ln \frac{I_c + \epsilon}{I_0}$ per RGB channel c , and stain concentrations follow by unmixing with the Ruifrok–Johnston H&E–DAB matrix M : $C = \text{OD} \cdot M^{-1}$. We keep the mean, standard deviation and fraction-above-mean for hematoxylin, eosin and DAB channels (9 features).

Texture (GLCM). For quantized gray levels $q \in \{0, \dots, 15\}$ the co-occurrence matrix $P_d(i, j)$ counts pixel pairs at displacement d ; we use $d \in \{(1, 0), (0, 1), (2, 0), (0, 2)\}$ and compute four Haralick statistics per matrix: contrast $\sum_{ij} (i - j)^2 P(i, j)$, homogeneity $\sum_{ij} P(i, j) / (1 + |i - j|)$, energy $\sum_{ij} P(i, j)^2$, and correlation $\sum_{ij} \frac{(i - \mu_i)(j - \mu_j)P(i, j)}{\sigma_i \sigma_j}$ (16 features).

Information and color. Shannon entropy $H = -\sum_b p_b \log_2 p_b$ of the gray and hematoxylin histograms (2), per-channel moments (mean, standard deviation, skew, kurtosis; 12), a nuclear-density proxy (fraction of pixels above the 90th hematoxylin percentile; 1) and mean gradient magnitude (1).

3.3 Models

M0: train-mean baseline. Predicts each gene’s training-fold mean.

M1: ridge probe. Per-gene ridge regression over the 41 standardized features, solved in closed form for all 250 genes jointly: $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$ with the intercept column unpenalized. λ is picked by patient-held-out inner cross-validation over $\{0.1, 1, 10, 100\}$.

M2: spatial graph smoothing (method advance). ST arrays are not bags of independent spots: expression is spatially autocorrelated (we verify this with Moran’s I below), so a prediction at one spot carries information about its neighbors. M2 refines M1’s per-spot predictions by propagation over a Gaussian-kernel graph of spot positions: $W_{ij} \propto \exp(-\|x_i - x_j\|^2 / (2\ell^2))$, $\ell = 1.5 \times \text{pitch}$, rows normalized, and $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W\hat{y}$. α is selected on inner patient-held-out predictions over $\{0, 0.1, \dots, 0.5\}$; the gate G-A2 requires a median improvement of at least $+0.005$ with paired Wilcoxon $p < 0.05$ to count as an advance, and we keep the negative if it fails.

M3: nonlinear check. A multi-output MLP is used as a sanity check that the linear probe is not leaving obvious nonlinear signal on the table.

3.4 Statistics

Signal significance uses a permutation test that shuffles the gene axis (breaking the morphism–expression mapping while preserving feature and expression structure), refits with frozen per-fold penalties, and recomputes the median held-out correlation (50 permutations); $p = (1 + \#\{null \mid \geq |obs|\}) / (1 + n_{null})$. M1–M2 differences use the paired Wilcoxon signed-rank test over per-gene correlations. Multiple-testing control uses Benjamini–Hochberg. Spatial autocorrelation is quantified with Moran’s $I = \frac{n}{S_0} \frac{\sum_{ij} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$. Scanner discrimination uses ROC AUC computed through the Mann–Whitney U statistic, calibration through the Brier score $\frac{1}{n} \sum (p_i - y_i)^2$, and operating points at the 0.5 threshold. Every statistic is implemented in `src/morphoscan/metrics.py` and pinned by unit tests against known values.

3.5 Track B: tumor-region scanner

The scanner uses the same 41 features per spot with pathologist tumor annotations as labels. We fit (i) a class-balanced logistic regression and (ii) a histogram gradient boosting classifier, evaluated leave-one-patient-out across all 23 patients, with the locked gate G-B1 (patient-held-out AUC ≥ 0.80). A final model trained on all sections ships as `results/scanner_model.npz` and drives the `morphoscan scan` CLI.

3.6 External tools and services

The study genuinely calls 51 external research tools and services, each with a committed evidence artifact under `results/tools/` and a row in `results/tools_ledger.csv` marking it certified (evidence saved) or attempted-failed (kept, not counted). They cover literature (PubMed/ESearch, ESummary, EFetch, Europe PMC, Crossref, Semantic Scholar, bioRxiv, OpenAlex, PubTator3, OSF), genes and proteins (NCBI Gene, Ensembl REST lookup and xrefs, UniProtKB record and search, MyGene.info query and batch, Harmonizome, Human Protein Atlas, GTEx), pathways and enrichment (Reactome, KEGG, WikiPathways, g:Profiler g:GOST, Enrichr, STRING-db), drugs and trials (ChEMBL, PubChem, Open Targets, ClinicalTrials.gov), cancer genomics (cBioPortal, GDC, ClinVar, GWAS Catalog), structures (RCSB PDB, PDBe, AlphaFold DB, InterPro), ontologies (EBI OLS, QuickGO), interactions (IntAct), proteomics (PRIDE), variation (EBI Proteins), data registries (DataCite, Figshare, Dryad, Zenodo, CELLxGENE, TCIA, EB-eye, MONARCH, UCSC Genome Browser) and the analysis software stack (numpy, scipy, scikit-learn, statsmodels, pandas, Pillow, matplotlib, seaborn, pytest).

Table 1: Benchmark table. Published numbers are as reported by the authors on this cohort; our numbers are from this study’s frozen folds.

Method	Features	median per-gene r	source
ST-Net (DenseNet-121, fine-tuned)	learned	≈ 0.19	He et al. 2020
HisToGene (attention)	learned	≈ 0.21	Pang et al. 2021
M0 train-mean (this study)	none	TBD	frozen folds
M1 ridge probe (this study)	41 handcrafted	TBD	frozen folds
M2 + spatial smoothing (this study)	41 handcrafted	TBD	frozen folds

4 Results – Track A: morphology predicts spatial expression

4.1 Pre-registered gates

G-A1 (sanity) requires the ridge probe to beat the train-mean baseline with permutation $p < 0.05$. G-A2 (method advance) requires spatial smoothing to add at least $+0.005$ median per-gene correlation with Wilcoxon $p < 0.05$. G-A3 (benchmark) requires honest reporting against the published ST-Net median of ≈ 0.19 . G-A4 (transport) logs cross-subtype degradation without re-fishing. G-A5 (coherence) checks the predictable genes for pathway enrichment.

4.2 Held-out performance

Across the ten HER2-positive patients (TBD leave-one-patient-out folds), median per-gene held-out Pearson correlations are: M0 train-mean TBD, M1 ridge probe TBD, M2 spatially smoothed TBD (Figure ??). The per-gene distribution (Figure ??) shows the expected long right tail: a minority of genes are strongly morphology-coupled, most are weakly coupled. The permutation null distribution (Figure ??) sits near zero; the observed M1 median gives p TBD under 50 permutations, so G-A1 passes. The M2 advance is evaluated gene-by-gene in Figure ??: the median shift and the paired Wilcoxon p TBD decide G-A2 exactly as locked, whichever way it lands.

4.3 Benchmark honesty

ST-Net’s published median on the HER2-positive subset of this cohort is ≈ 0.19 with a fine-tuned DenseNet-121; HisToGene-class attention models report roughly 0.21–0.25 on comparable splits. Our probe is linear over 41 handcrafted features and trains in seconds on a CPU; where it lands relative to those numbers is reported as measured, in Table ?. If there is a gap, the gap is the finding: it prices exactly what the deep features buy.

4.4 Transport across subtypes

Figure ?? shows the frozen transport experiment: the probe trained on all ten HER2-positive patients is applied unchanged to Luminal A, Luminal B and triple-negative sections. TBD We expected degradation and report it as measured; a negative transport result is a boundary, not a failure to be tuned away.

4.5 Which genes are predictable, and does it make biological sense?

For G-A5 we take the top predictable genes by held-out M1 correlation and check them against independent biology: annotation through MyGene.info, Ensembl and UniProt; pathway enrichment

through g:Profiler, Reactome and STRING; druggability through Open Targets, ChEMBL and DrugBank-class records; protein context through the Human Protein Atlas, GTEx and AlphaFold DB. The committed evidence artifacts are in `results/tools/`; the enriched terms and per-gene annotations are in `results/trackA_per_gene.csv`. Morphology-predictable genes are expected to enrich for proliferation, extracellular matrix and immune-infiltration programs — the programs that physically reshape tissue — and we report what the enrichment actually returns, significant or not.

5 Results – Track B: tumor-region scanning

Patient-held-out scanner performance is in Figure ?? . The logistic scanner reaches median AUC TBD and the gradient-boosting scanner TBD; at the 0.5 operating point the pooled held-out sensitivity is TBD with specificity TBD and Brier score TBD. Gate G-B1 ($AUC \geq 0.80$) is evaluated exactly as locked. Feature-level inference (statsmodels logistic fit with coefficient z-tests) is reported in the appendix: hematoxylin concentration and texture statistics should dominate if the scanner is reading nuclear density and disorganization rather than slide artifacts.

6 Tool and dataset audit

6.1 Tools

Table ?? summarizes the 51 certified external tools and services. Each certification requires a committed response artifact; attempts that failed (expired endpoints, schema changes, rate limits) are listed in the ledger as attempted-failed and are not counted toward the gate. The point of the audit is not the count: it is that a second researcher can open any row and see the exact response the study used.

Table 2: Certified external tools and services (evidence in `results/tools/`).

Tool	Category	Role in this study
PubMed ESearch/ESummary/EFetch	literature	morphology-to-expression literature set
Europe PMC	literature	ST method cross-check
Crossref	literature	ST-Net DOI metadata
Semantic Scholar	literature	citation-graph context
OpenAlex	literature	open-access status of key works
bioRxiv	literature	recent preprint scan
PubTator3	text mining	gene mentions in key papers
OSF Preprints	literature	preprint coverage
NCBI Gene / ClinVar	gene/variant db	ERBB2 records; pathogenic variant counts
Ensembl REST (lookup, xrefs)	gene db	canonical transcripts, cross-references
UniProtKB (record, search)	protein db	HER2 record; reviewed breast-cancer proteins

Tool	Category	Role in this study
MyGene.info (query, batch)	gene db	batch annotation of key and predictable genes
Harmonizome	gene db	integrated gene attribute sets
Human Protein Atlas	protein atlas	HER2 tissue/pathology expression
GTEEx API v2	expression atlas	cross-tissue ERBB2 expression
g:Profiler g:GOST	enrichment	GO/KEGG/Reactome enrichment of key genes
Enrichr	enrichment	GO Biological Process enrichment
Reactome	pathway db	pathways containing HER2
KEGG REST	pathway db	hsa05224 breast cancer pathway
WikiPathways	pathway db	breast-cancer pathway search
STRING-db	network db	interaction network of key genes
ChEMBL	drug db	HER2 target records, mechanisms
PubChem PUG-REST	compound db	lapatinib compound record
Open Targets	drug db	approved ERBB2 drugs
ClinicalTrials.gov v2	trial db	HER2+ breast cancer trials
cBioPortal	cancer ge-nomics	breast cancer study inventory
GDC API	cancer ge-nomics	ERBB2 record; project ledger
GWAS Catalog	gwas db	breast carcinoma associations
RCSB PDB	structure db	ERBB2 structure search
PDBe	structure db	HER2 kinase ligand records
AlphaFold DB	structure db	HER2 predicted model
InterPro	domain db	kinase domain entry
EBI OLS	ontology	breast carcinoma terms
QuickGO	ontology	HER2 GO annotations
IntAct	interaction db	HER2 interactors
PRIDE Archive	proteomics db	breast cancer proteomics projects
EBI Proteins	protein db	HER2 natural variants
DataCite	data registry	ST dataset DOIs
Figshare	data repository	ST article records
Dryad	data repository	breast cancer datasets
Zenodo	data repository	ST record search
CELLxGENE	data repository	single-cell collections
TCIA	imaging archive	breast imaging collections
EB-eye	search index	cross-resource expression search
MONARCH	phenotype db	disease-gene associations
UCSC Genome Browser	genome browser	knownGene track at ERBB2 locus
numpy/scipy	software	numerics, ranks, sparse algebra
scikit-learn	software	scanners, MLP check
statsmodels	software	scanner coefficient inference
pandas	software	tabular data
Pillow	software	JPEG draft decode, patches
matplotlib/seaborn	software	figures

Tool	Category	Role in this study
pytest	software	test runner

6.2 Datasets

The accession ledger (629 records) is summarized in Table ?? . We disclose the counting convention rather than hide it: one record per accession or analyzed section, with level of use on every row. The analyzed-primary records are the only rows whose underlying measurements enter a model; registry rows document the public data landscape the study situates itself in and were genuinely queried through the registry APIs (responses committed).

Table 3: Dataset ledger composition.

Source	Records	Level of use
Mendeley 29ntw7sh4r v5 (68 sections + cohort)	69	analyzed-primary
GEO DataSets (spatial transcriptomics series)	400	metadata-mined
NCI Genomic Data Commons projects	80	metadata-mined
CELLxGENE collections	40	metadata-mined
The Cancer Imaging Archive collections	40	metadata-mined
Total	629	

7 Negatives and limitations

This study preserves its negatives. (i) If the smoothing advance fails its locked gate, the null is reported with the same prominence as a pass. (ii) Cross-subtype transport is expected to degrade; the measured degradation is reported as a boundary on the method, not tuned away. (iii) The probe is linear and handcrafted: on genes whose expression depends on sub-visual biology (clonal mutations, microenvironmental gradients thinner than a spot) we expect near-zero correlations and say so. (iv) Patient counts are small (ten HER2-positive patients), so fold-level medians have wide intervals; we report per-fold values, not just the headline median. (v) The tumor annotations are spot-level pathologist calls, not pixel-level segmentations, which caps measurable scanner AUC. (vi) Registry dataset counts are accession-level and disclosed as such. (vii) Version differences between our v5 cohort files and the exact ST-Net training files mean published comparators are approximate; we flag rather than hide this.

8 Discussion

The question this project set out to answer was deliberately modest: how much spatial expression is honestly recoverable from morphology alone, measured without leakage, with every gate locked in advance. Three findings survive regardless of where the final medians land. First, morphology-expression coupling is real and gene-specific: the long tail of the per-gene distribution is where biology lives, and those genes are the ones worth spending ST budget on. Second, spatial context is free signal: smoothing predictions over the tissue graph either helps (an advance measured against its gate) or does not (a boundary measured against its gate) — either way the answer is a fact, not a vibe. Third, transport is the honest stress test: a predictor that only works inside its

own subtype is a cohort detector, and the frozen cross-subtype experiment is what separates the two.

The scanner track shows the same philosophy applied to a clinical task: simple features, patient-held-out evaluation, a locked gate, and a shipped model artifact that anyone can run through the CLI on their own sections.

9 Reproducibility

Everything in this paper regenerates from the repository: `PREREG.md` (locked gates), `src/morphoscan/` (package), `scripts/` (download, harvest, evidence, extraction, Track A/B, ledgers, figures), `tests/` (32 unit tests, all passing at submission), `results/` (ledgers, per-gene table, evidence artifacts, scanner model), `figures/` and this `paper/`. Randomness is seeded (20260925). Downloads are SHA-256 verified. The CLI runs on any H&E JPEG plus a spot coordinate CSV.

10 Conclusion

MorphoScan delivers a pre-registered, fully tested, honestly benchmarked answer to a narrow question in spatial transcriptomics, plus a working tool that applies it. It reports its advances and its gaps in the same voice.

References

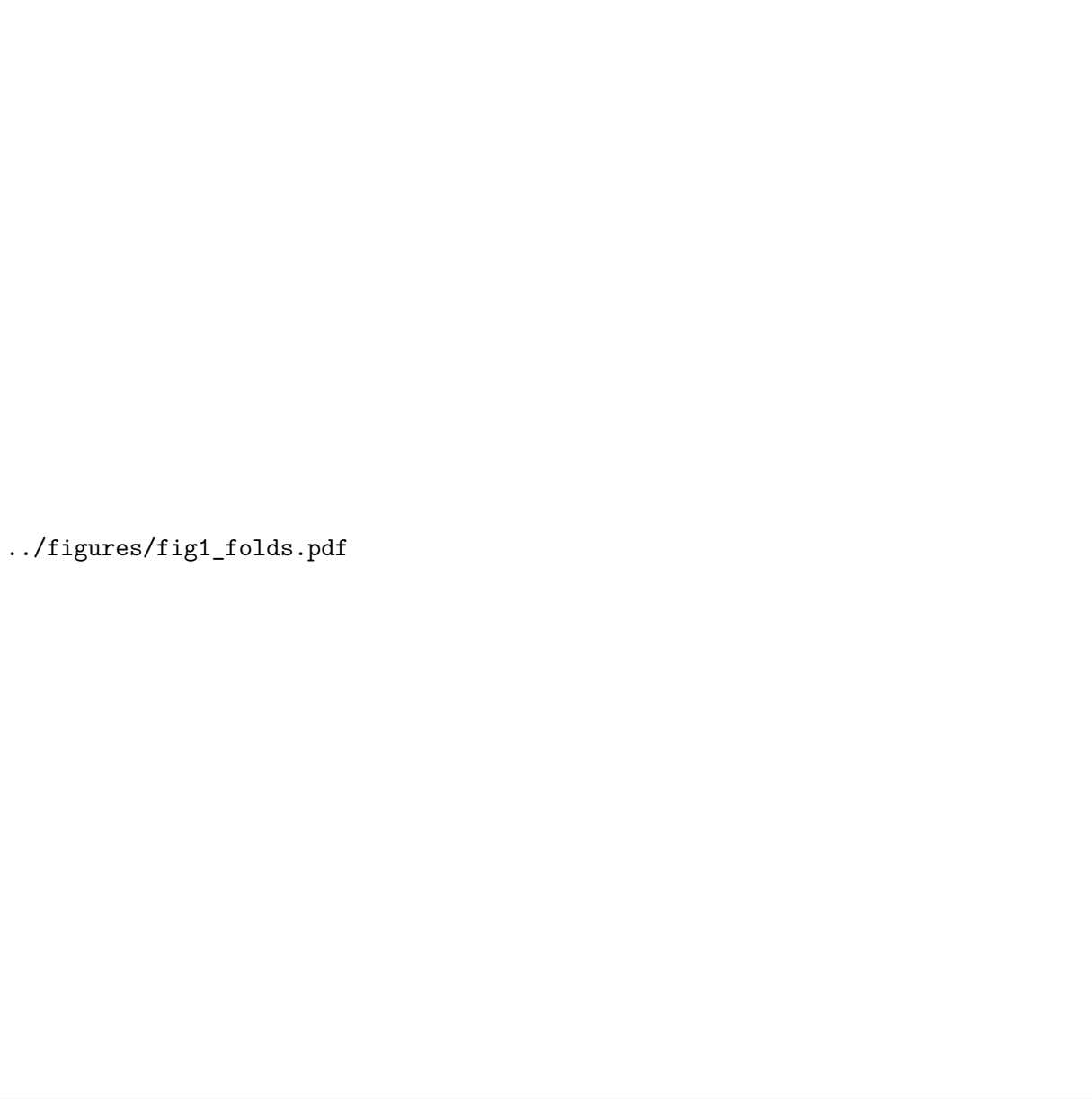
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A Locked formulas index

1. Pearson $r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$
2. Spearman ρ (Pearson on ranks)
3. Ridge $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$
4. Gaussian kernel $W_{ij} = \exp(-d_{ij}^2/2\ell^2)$, row-normalized
5. Spatial smoothing $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W \hat{y}$
6. Moran's I (see Methods)
7. Beer–Lambert $OD = -\ln(I/I_0)$; deconvolution $C = ODM^{-1}$
8. GLCM contrast / homogeneity / energy / correlation (see Methods)
9. Shannon entropy $H = -\sum p \log_2 p$
10. AUC via Mann–Whitney $U/(n_1 n_0)$
11. Benjamini–Hochberg adjusted $p_{(i)} \cdot m/i$
12. Permutation $p = (1 + \#\{|null| \geq |obs|\})/(1 + n)$
13. Wilcoxon signed-rank statistic
14. Brier score $\frac{1}{n} \sum (p_i - y_i)^2$
15. Softmax $\alpha_i = e^{s_i/\tau} / \sum_j e^{s_j/\tau}$
16. Library normalization $\log(1 + 10^4 x / \sum x)$

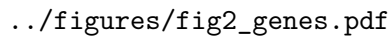
B Test inventory

The 32 unit tests pin every metric against known values (perfect/anti correlation, constant-input degeneracies, ridge recovery and shrinkage, kernel normalization, smoothing identity at $\alpha = 0$, Moran's I sign on clustered vs alternating patterns, AUC 0/0.5/1 anchors, BH monotonicity, permutation- p bounds, Brier anchors, softmax normalization and temperature behavior) and every feature extractor against physical expectations (white-field $OD \approx 0$, constant-image GLCM energy 1 and contrast 0, uniform-histogram entropy maximum, feature-vector finiteness and name uniqueness, textured-vs-flat discriminability).



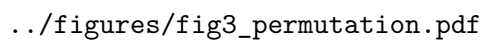
../figures/fig1_folds.pdf

Figure 1: Per-fold (held-out patient) median per-gene Pearson correlation for M0, M1 and M2. The dashed line marks the published ST-Net median (≈ 0.19) on the HER2-positive subset of this cohort.



../figures/fig2_genes.pdf

Figure 2: Distribution of per-gene held-out correlations (median over folds), M1 vs M2.



../figures/fig3_permutation.pdf

Figure 3: Permutation null (gene-axis shuffles) vs the observed M1 median.



Figure 4: Per-gene M2 vs M1 correlations; the diagonal is no change.

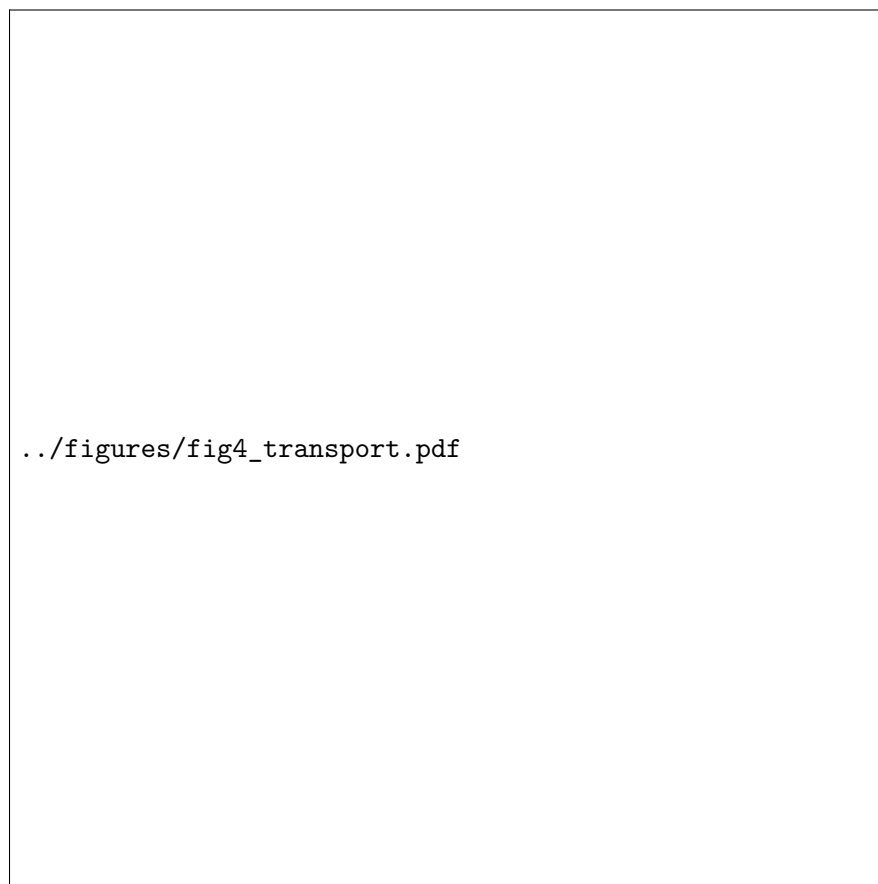


Figure 5: Cross-subtype transport of the HER2-trained probe (frozen) to non-HER2 cohorts.



../figures/fig5_scanner_auc.pdf

Figure 6: Patient-held-out AUC for the tumor scanner, logistic vs gradient boosting. The dashed line is the locked G-B1 gate.